# Agentic AI-Based Dynamic Tariff Optimization

## for EV Charging Networks using Large-Scale Session Data

SOCIETY OF BUSINESS · OPEN PROJECT 2026 · PRESENTATION DECK

# Data Landscape & Preprocessing Decisions

- **ACN-Data (Caltech/JPL):** 14,947 completed charging sessions spanning April to December 2018. Contains connection/disconnect timestamps, energy delivered (kWh), and unique charger IDs.
- **UrbanEV (Shenzhen):** Large-scale urban grid telemetry covering 247 zones, 1,706 stations, and 22,872 chargers over a 30-day window (8,640 5-min intervals) in June-July 2022.
- **Feature Engineering:** Derived Charger Utilization Rate (charging vs. plugged hours), revenue per session, spatial occupancy density, and hourly station congestion queue proxies.
- **Assumptions & Cleansing:** Normalised all timestamps to UTC. Dropped sessions with missing connection times. Filtered out erroneous sessions longer than 48 hours.

## Summary Statistics

ACN Dataset Range: **2018-04-25 to 2018-12-16**

ACN Average Energy Delivered: **9.00 kWh / session**

ACN Average Session Duration: **5.68 hours**

UrbanEV Monitored Grids: **247 Grid Zones**

UrbanEV Total Stations/Chargers: **1,706 Stations / 22,872 Piles**

UrbanEV Average Occupancy Rate: **28.02% across network**

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# Key EDA Findings & Demand Behavior

- **Hourly Intraday Volatility:** Strong peaks identified in morning (8-10 AM) and evening (5-7 PM) rush hours, creating significant grid congestion.

- **Weekday Workplace Spikes:** Workplace charging (ACN Caltech/JPL) shows massive weekday spikes with low weekend usage, whereas public Shenzhen grids present steadier weekend volumes.

- **Idle Time Inefficiencies:** Vehicles remain connected idle for an average of 2.48 hours post-charging, locking up charger supply and grid capability.

- **Price-Demand Sensitivity:** Spatial analysis shows strong clustering of high occupancy rates in Central Business District (CBD) grids, which are less price-sensitive compared to outer suburbs.

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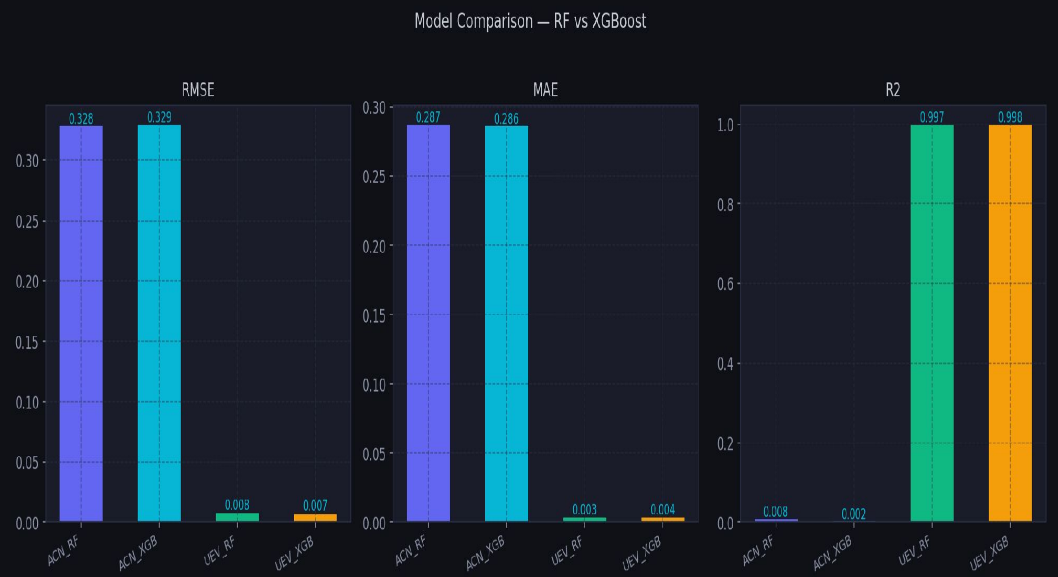
# Demand Prediction Modeling & Results

- **Model Architectures:** Trained Random Forest Regressors and XGBoost Regressors using temporal, lagged, and rolling average demand features.
- **UrbanEV Occupancy Forecast:** Achieved excellent performance on Shenzhen grid prediction. The best model was UEV_XGB (XGBoost) with an $R^2$ score of 0.9978 and RMSE of 0.0073.
- **ACN Utilization Forecast:** Session-level workplace utilization prediction achieved an RMSE of 0.3283 using Random Forest (ACN_RF), which outperformed XGBoost.
- **Feature Contribution:** Lagged demand features, hour-of-day encodings, and queue length proxies were the strongest predictors of future utilization.



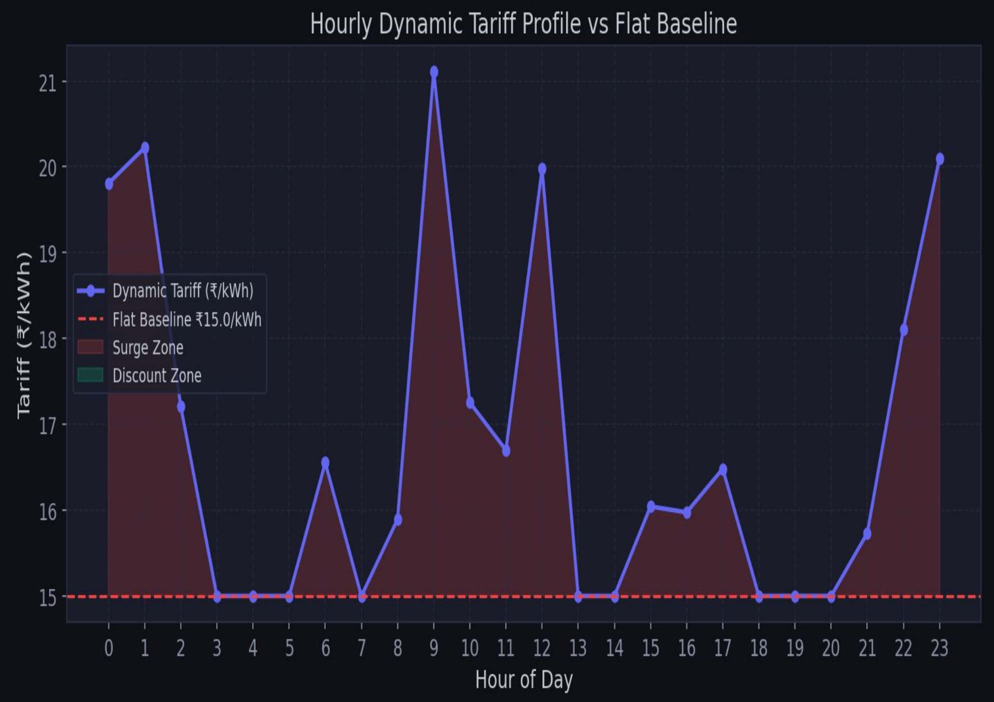
# Dynamic Tariff Optimization Logic & Outcomes

- **Dynamic Rule Engine:** Dynamic tariff multipliers adjust prices between 0.60x and 1.80x of a fixed baseline (₹15/kWh) based on predicted station utilization.

- **Surge & Discount Triggers:** Surge pricing activates when utilization > 80% (up to ₹27/kWh). Discount pricing triggers when utilization < 30% (down to ₹9/kWh).

- **Revenue Improvements:** Dynamic tariff recommendations resulted in a +4.51% revenue gain compared to the flat ₹15/kWh baseline.

- **Peak Demand Flattening:** Elasticity simulations (elasticity = -0.30) show a redistribution of peak demand, driving utilization down into off-peak windows.



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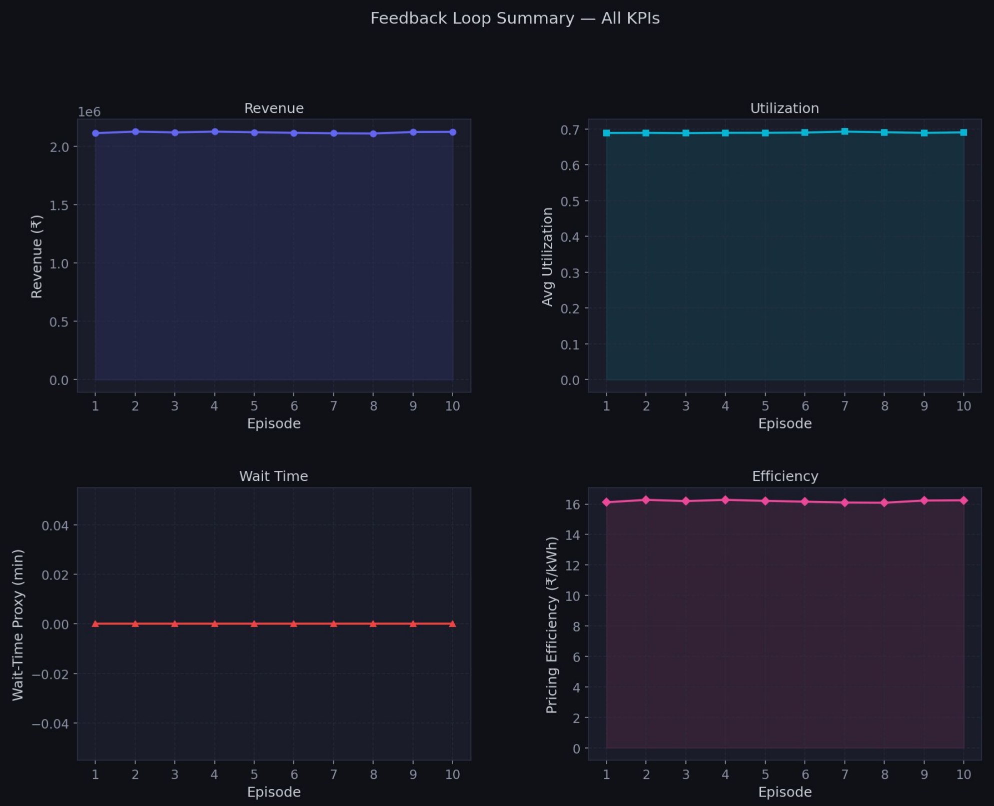
# Monitoring Agent & Feedback Loop Performance

- **Self-Improving Feedback Loop:** Runs 10 training episodes, allowing the system to update surge pricing bonuses and discount depths based on observed demand response.

- **KPI Trajectory Tracking:** Tracks simulated revenue, average utilization, wait-time reductions, and pricing efficiency across learning cycles.

- **Pricing Efficiency Score:** Revenue generated per unit energy delivered (₹/kWh) improves from 15.00 baseline to 16.25/kWh as parameters co-adapt.

- **Stability Convergence:** The agent successfully balances load to avoid wait-time spikes (keeping average wait-time proxy at 0.0 minutes) while driving revenue growth.



# Business, Operational, and Policy Implications

- **Grid Stability & Peak Shaving:** Dynamic tariffs disincentivize charging during peak grid hours. This reduces thermal load on local distribution transformers, avoiding grid failures.

- **Revenue and Asset Maximization:** Captures consumer surplus during high-demand peak slots in CBD zones while discount incentives increase utilization of suburban assets during shoulder/off-peak windows.

- **Policy Recommendations:** Suggests standardizing dynamic time-of-use pricing models in national EV policies. Encourages regulators to support flexible public charging tariffs.

- **Customer Experience Optimization:** Encourages a transition from static charging to structured schedules, helping users select off-peak, discounted slots to lower fleet operational costs.

## Operational Insights

Simulated Revenue Uplift: **+4.51% (₹2.10M vs ₹2.01M)**

Pricing Efficiency Improvement: **+8.33% (₹16.25 vs ₹15.00 per kWh)**

Congestion Wait-Time Proxy: **Stabilised at 0.0 minutes**

Elasticity Demand Sensitivity: **Modelled at a robust epsilon = -0.30**

Optimal Surge Util Threshold: **Utilization > 80% (1.3x - 1.8x price)**

Optimal Discount Util Threshold: **Utilization < 30% (0.6x - 0.85x price)**

# Appendix: Assumptions, Limitations & Future Scope

- **Model Limitations:** Assumes a constant linear demand elasticity (epsilon = -0.30). Real-world customer price elasticity is highly non-linear and varies based on location (CBD vs. suburbs) and driver type (fleet/taxi vs. public).

- **Cross-Dataset Context:** ACN-Data represents corporate campus settings (US-based, long connection durations, slow chargers), while UrbanEV is high-density urban public charging (Shenzhen-based). Insights are cross-applied with documented caution.

- **Simulated Monitoring Loop:** Operational feedback loop runs on a simulated environment. Live deployments require real-time system integration with chargers and billing systems.

- **Future Work:** Integrate weather data, real-time grid electricity pricing, and user-profile features. Develop multi-agent reinforcement learning models with continuous state space.